Research Methods (TP)

Practice: Research Methodology and Research Literature

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Activity I: Draft of your Research Question

Write your research question(s)

- What do you want to research in the form of 1-2 questions that capture the work's essence

Briefly think about

- How will you answer your research questions (selecting case/artifact, data collection) ?
- How will you analyze the data ?
- What results do you expect to obtain?

- **Optional**: Sketch an Illustration of your research process

- Discuss with the instructor

Libraries and Online Literature Databases

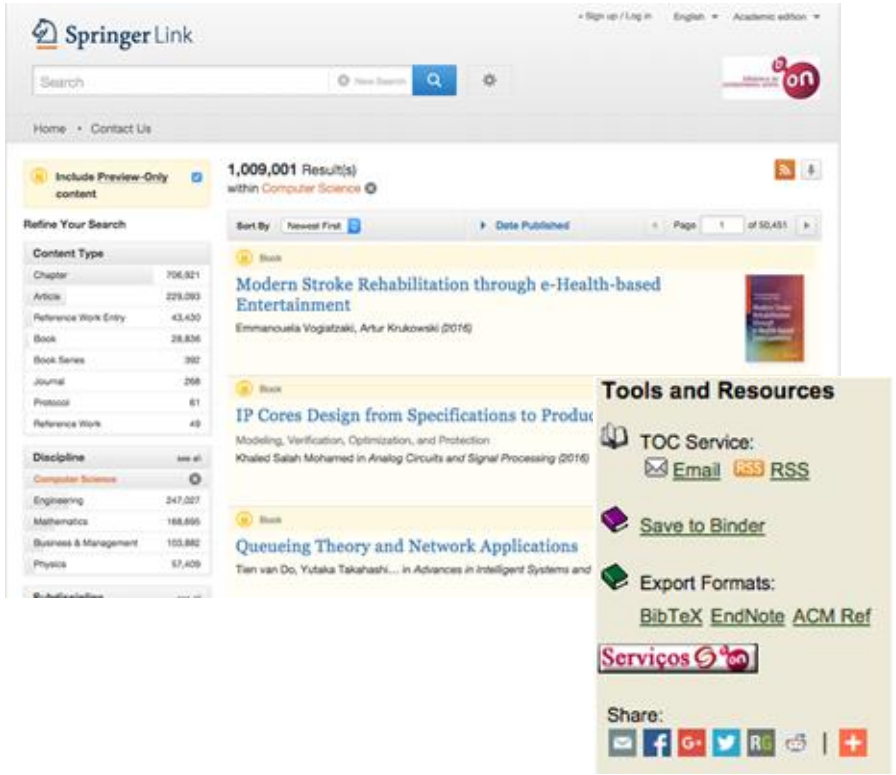
Academic libraries focus on organizing information to meet the needs of academics

Organized by, e.g.:

- established fields of research, types of publication (books, thesis, journals,…)

Online Libraries

- Scientific Repository of the Polytechnic Institute of Porto
  - https://recipp.ipp.pt/
  - DEI MSc Dissertations: https://recipp.ipp.pt/handle/10400.22/364
- B-On
  - http://www.b-on.pt/



ACM Digital Library (Pay walled; B-On)

- http://dl.acm.org/

Examples of Relevant Online Libraries/Databases

ACM Digital Library (Pay walled; B-On)

- http://dl.acm.org/

IEEE Explore (Pay walled; B-On)

- http://ieeexplore.ieee.org/

DBLP

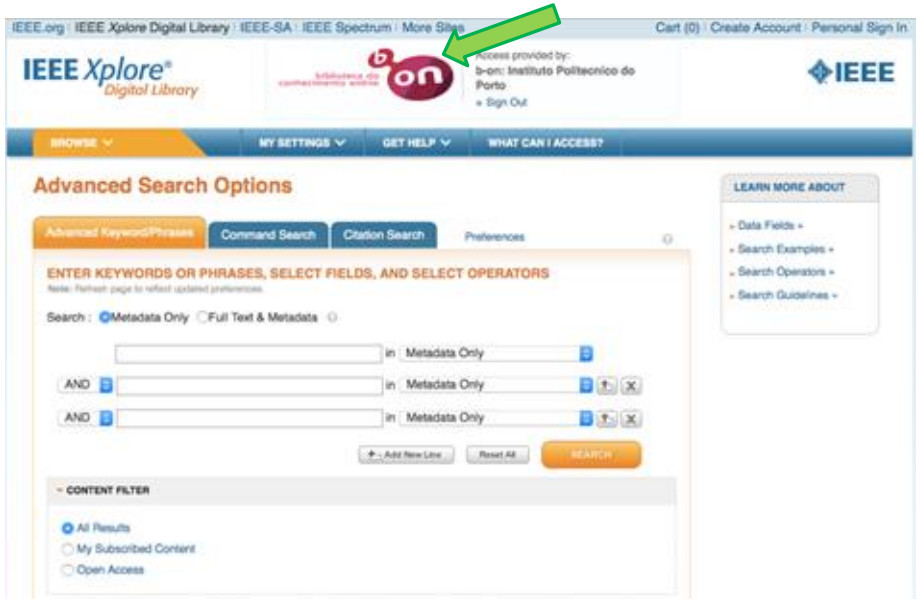
- http://dblp.uni-trier.de/ (Free)

The Collection of Computer Science Bibliographies (Free)

- http://liinwww.ira.uka.de/bibliography/

Arxiv.org (Free)

- http://arxiv.org/



Recording and Organizing Information

Always record your findings adequately

- Record:
  - author(s),
  - title,
  - year of publication,
  - publisher,
  - edition/version/issue/volume,
  - access information (URL, DOI, Reference ID…),
  - location,
  - abstract, keywords (preferably)…

Don’t trust your memory!…

Many different ways:

- Cards with notes (physically or on the computer)
- Articles (physically or on the computer) with text highlighted and notes
- **Reference management software**



Reference Management Software

There are many tools to help you collect, organize, cite, and share your research sources

Zotero
- free reference manager
- https://www.zotero.org/

Mendeley
- free reference manager and academic social network
- https://www.mendeley.com/

EndNote
- commercial reference manager
- http://endnote.com/

Use one!







Activity II: Search Digital Libraries

Develop the suggested activity (see ”Activity II: Search Digital Libraries” in Moodle):

1. Search the Acm Digital Library
2. Search the SciMago Journal & Country Rank (JSR)
3. Search the Web of Science (WoS)

Summary

Drafting your Research Question

Tools for searching and managing your references